Milestone 1 – Idea presentation
Feb 19 Th

If you have formed your group, put your name on the ELMS->People->Group->Semester-long Project

Documentation Due Feb 22 Sun

Groups (15)

Semester long project group 1	0 / 5 students
Semester long project group 2	0 / 5 students
Semester long project group 3	0 / 5 students
Semester long project group 4	0 / 5 students
Semester long project group 5	0 / 5 students
Semester long project group 6	0 / 5 students
Semester long project group 7	0 / 5 students
Semester long project group 8	0 / 5 students
Semester long project group 9	0 / 5 students
Semester long project group 10	0 / 5 students

5 min presentation + 3 min Q& A

2 Options:

a) Haven’t decided on the idea:

Present 3 of your best ideas and explain to us with sketches

b) Know what to do:

Present your final idea – what are the functions, challenges and potential solutions

5 min presentation + 3 min Q& A

Submit a google doc with:

a) Problem Statement & Idea
b) System Block Diagram
c) Input/Sensing+Output/Actuation
d) Challenges + Potential Solutions
e) Bill of Materials(BOM)

Braille Label Bot

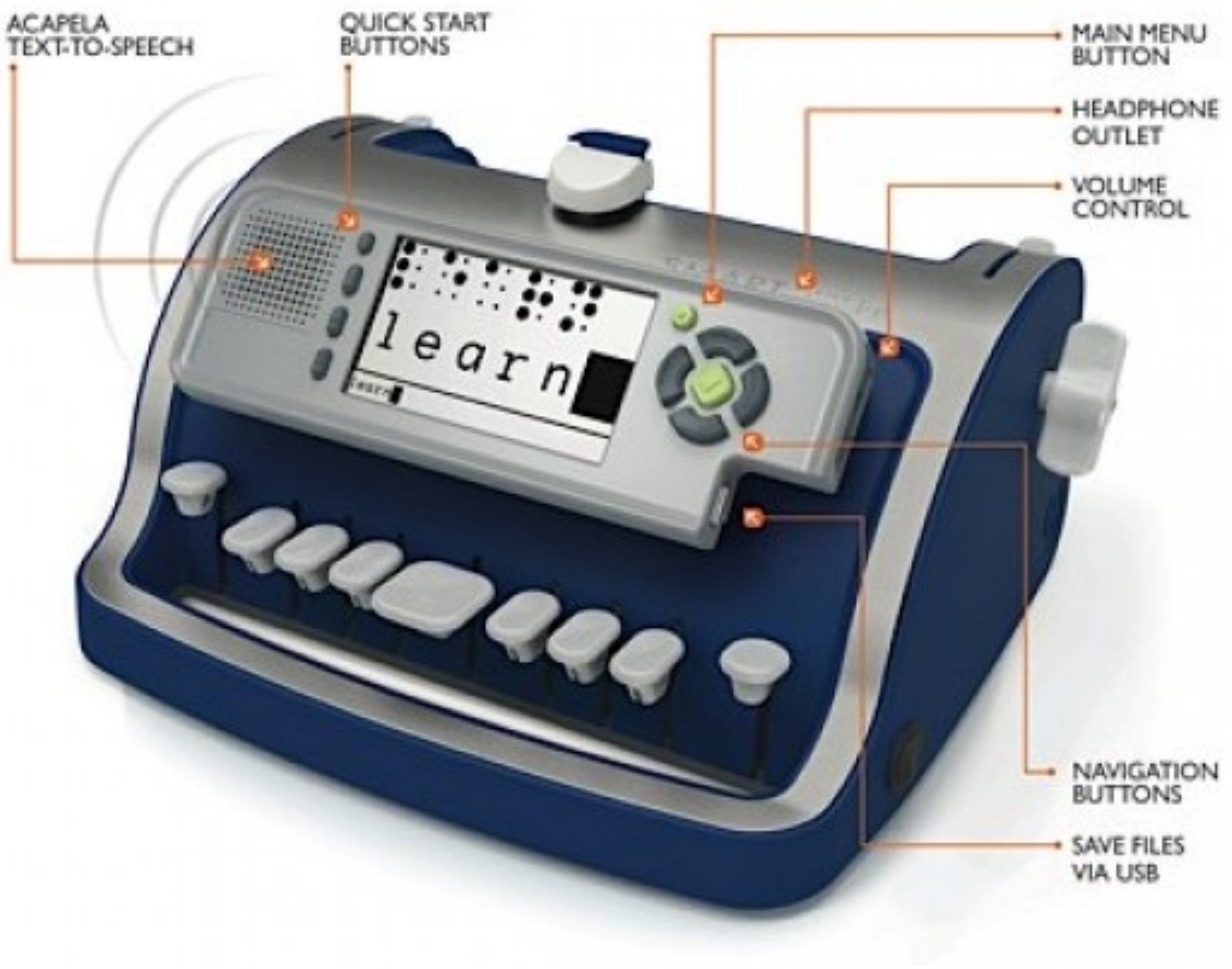
Huaishu


label.m
PROFESSIONAL HAIRCARE
Honey & Oat Shampoo/750 ml
label.m
PROFESSIONAL HAIRCARE
Honey & Oat Conditioner/750 ml

SHOPPERS DRUG MART
PROTEA PHARMACY INC.
210 WALLACE AVE. N. LISTOWEL, ON N4W 1K9 CANADA
Store # 1007
Rx#9373889 Ref:0
EASTDALE PUBLIC SCHOOL
TAKE 1 TABLET TWICE DAILY
TAKE WITH FOOD
MEDICINE
50 TAB
27 May 2017
HOPPERS DRUG MART
PROTEA PHARMACY INC.
210 WALLACE AVE. N. LISTOWEL, ON N4W 1K9
Store # 1007
9373892 Ref:0
TDALE PUBLIC SCHOOL
3 TIMES DAILY FOR 7 DAYS
AND KEEP IN REFRIGERATOR
EDICINE
27 May 2017

CONFERENCE RO

Perkins SMART Brailler



$ 2K+

6 Dot Braille Label Maker

$ 775

Reizen Braille Labeler

$ 40+
But fully manual
Hard to use

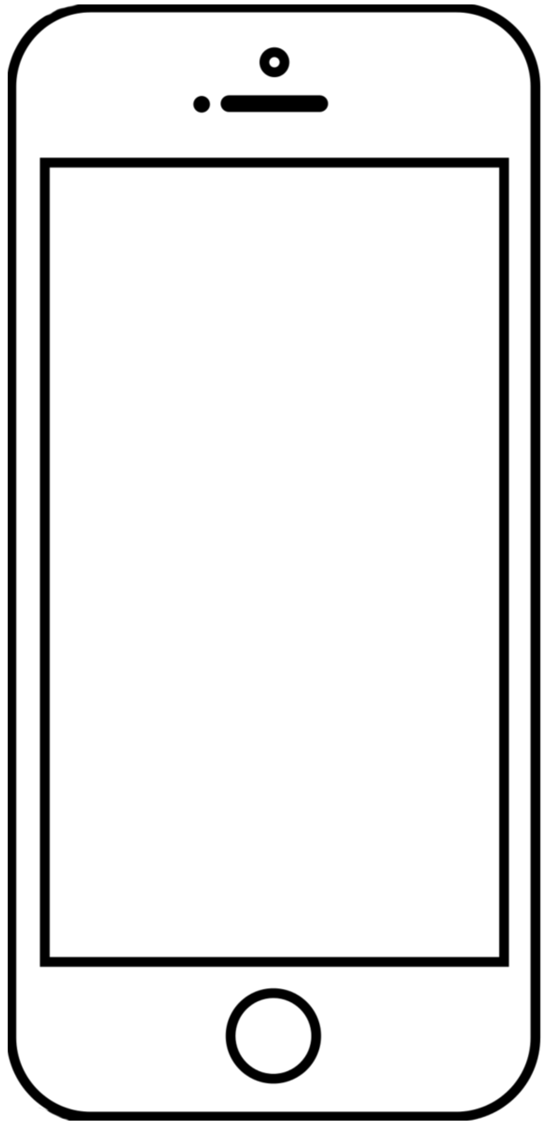
The idea

Low-cost, portable braille label bot that can be used by everyone

Can be used both
Indoor and outdoor

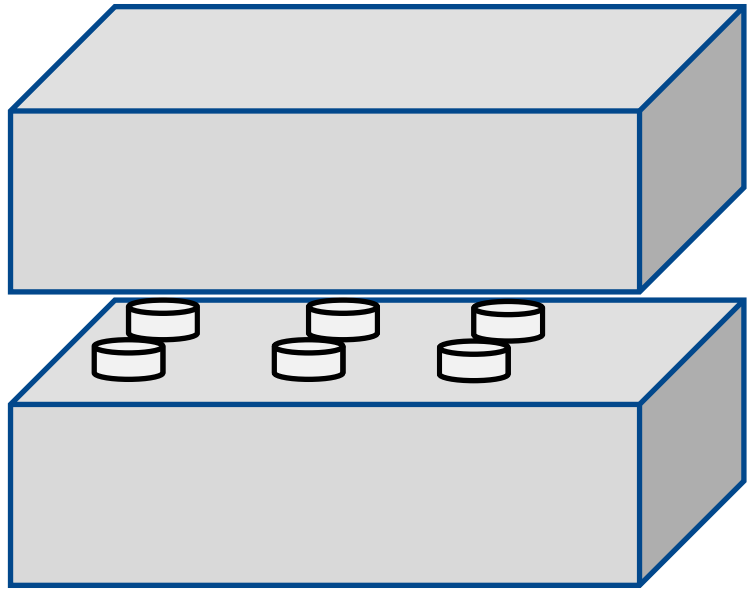
We can create braille
label for them

How



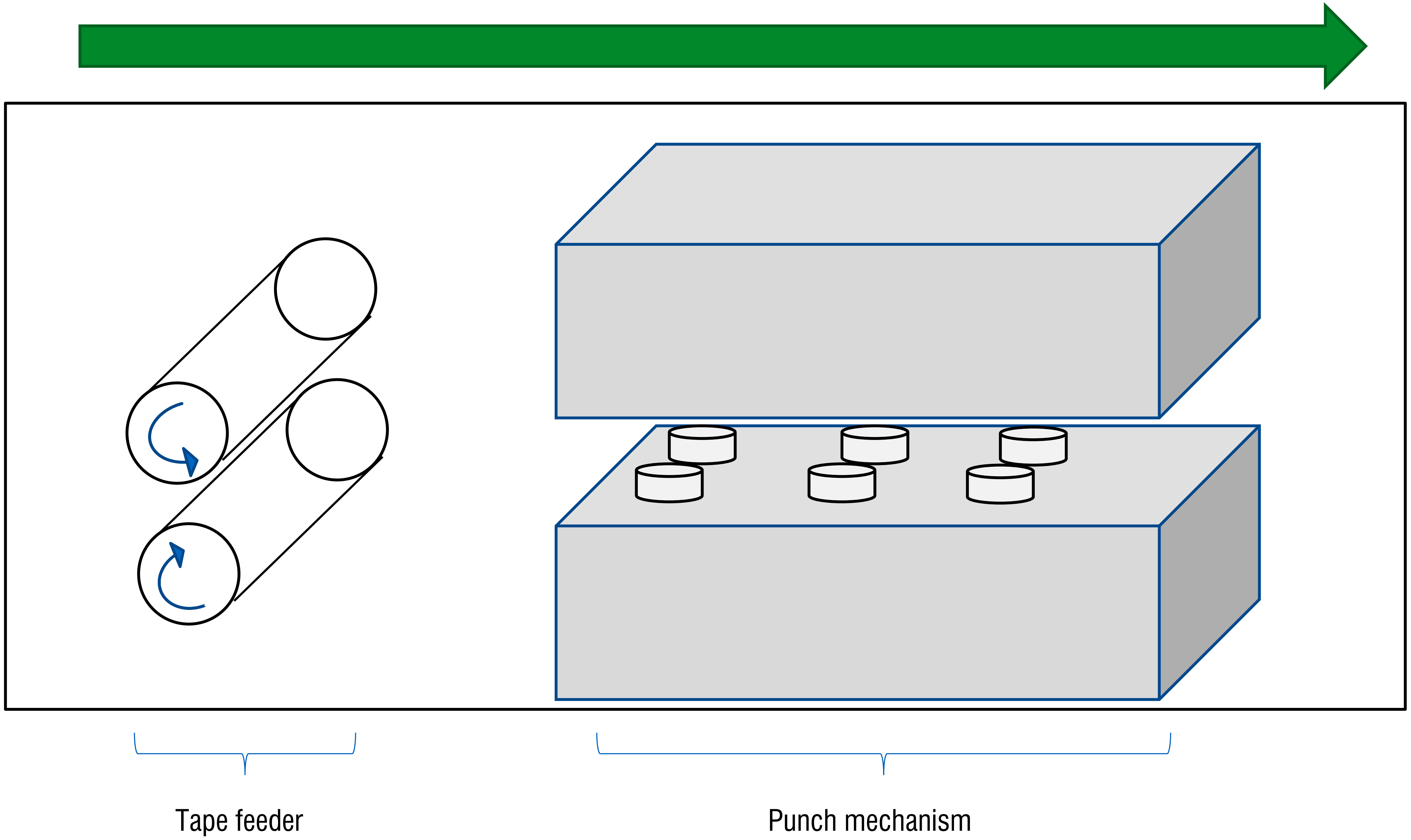
input platform
Either voice or type

+

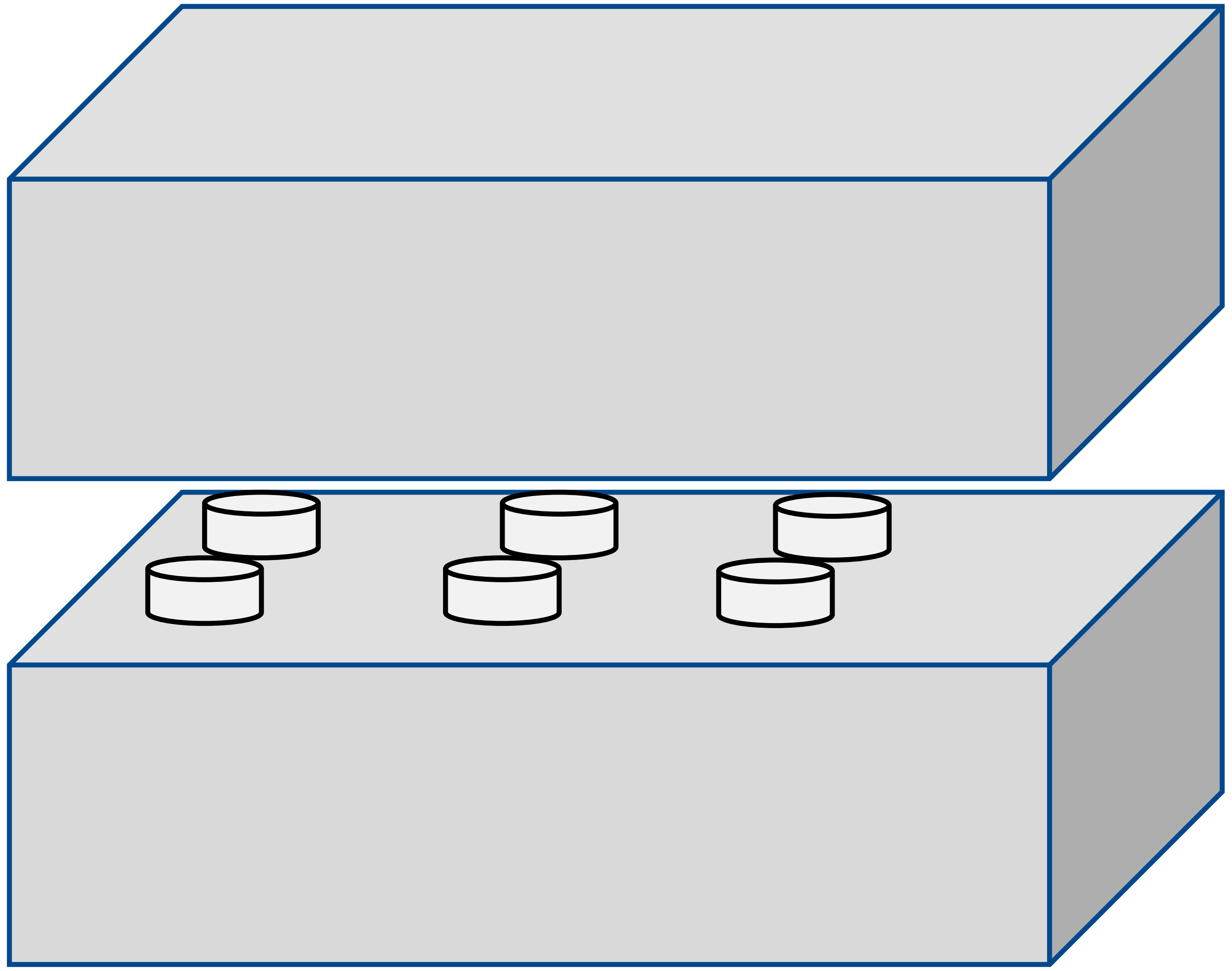


custom build machine
6 pin controlled with Arduino
punch holes to tape

How



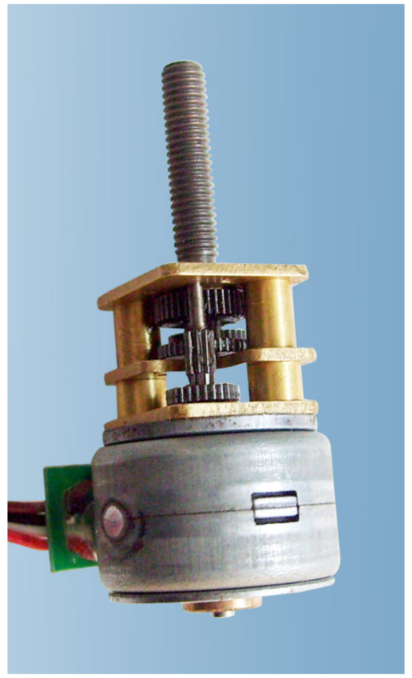
Main challenge



1. Limited physical space, dots need to be close to each other – how to arrange motors to control each of the 6 pins

2. Need large force to create hole or embossing

Potential solutions



Plan for the next milestone

1. Figure out the motor to create embossing

2. Create one working prototype that can create 2 dots at a close distance

ClawMyDesk

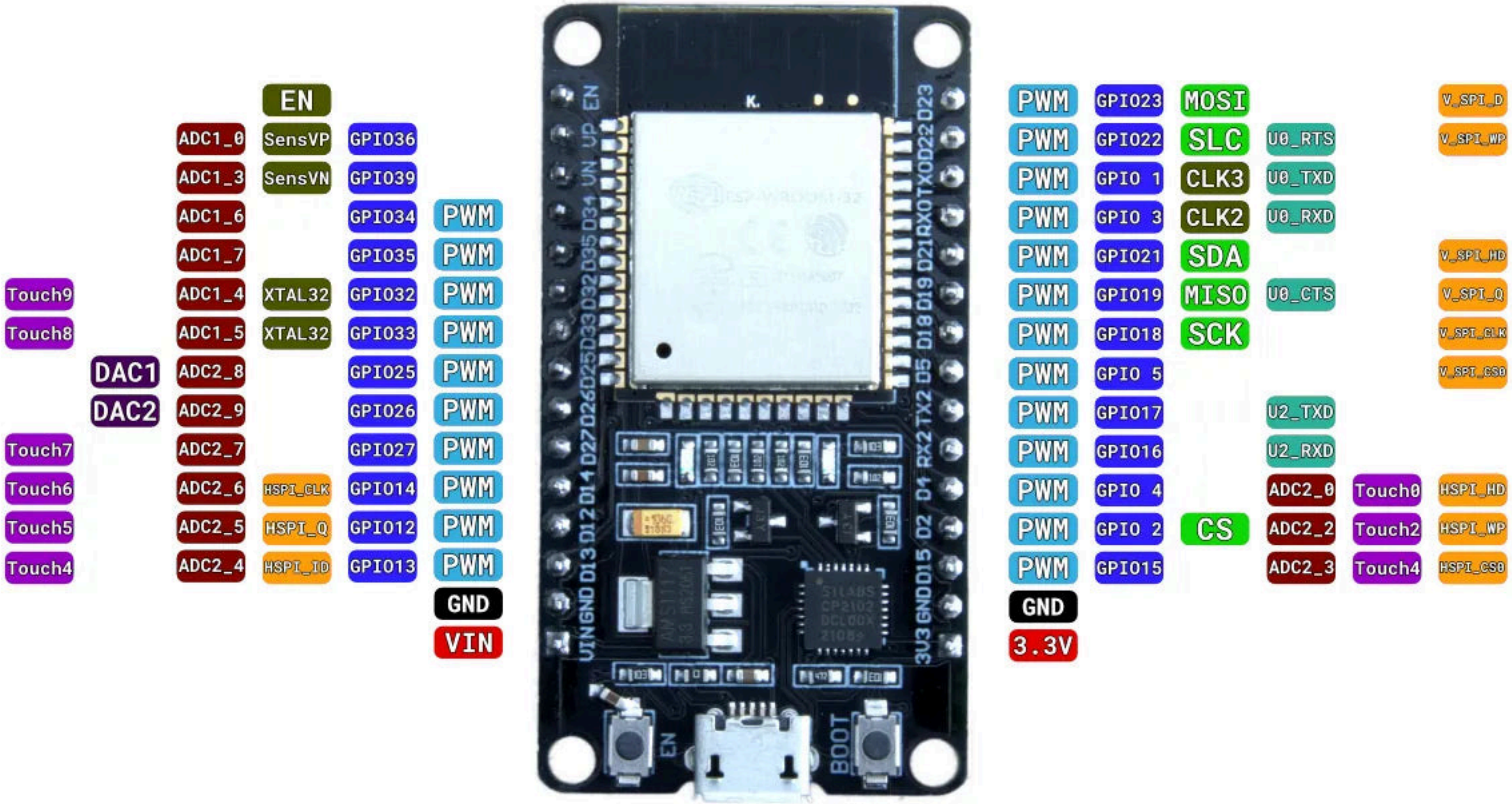
Huaishu

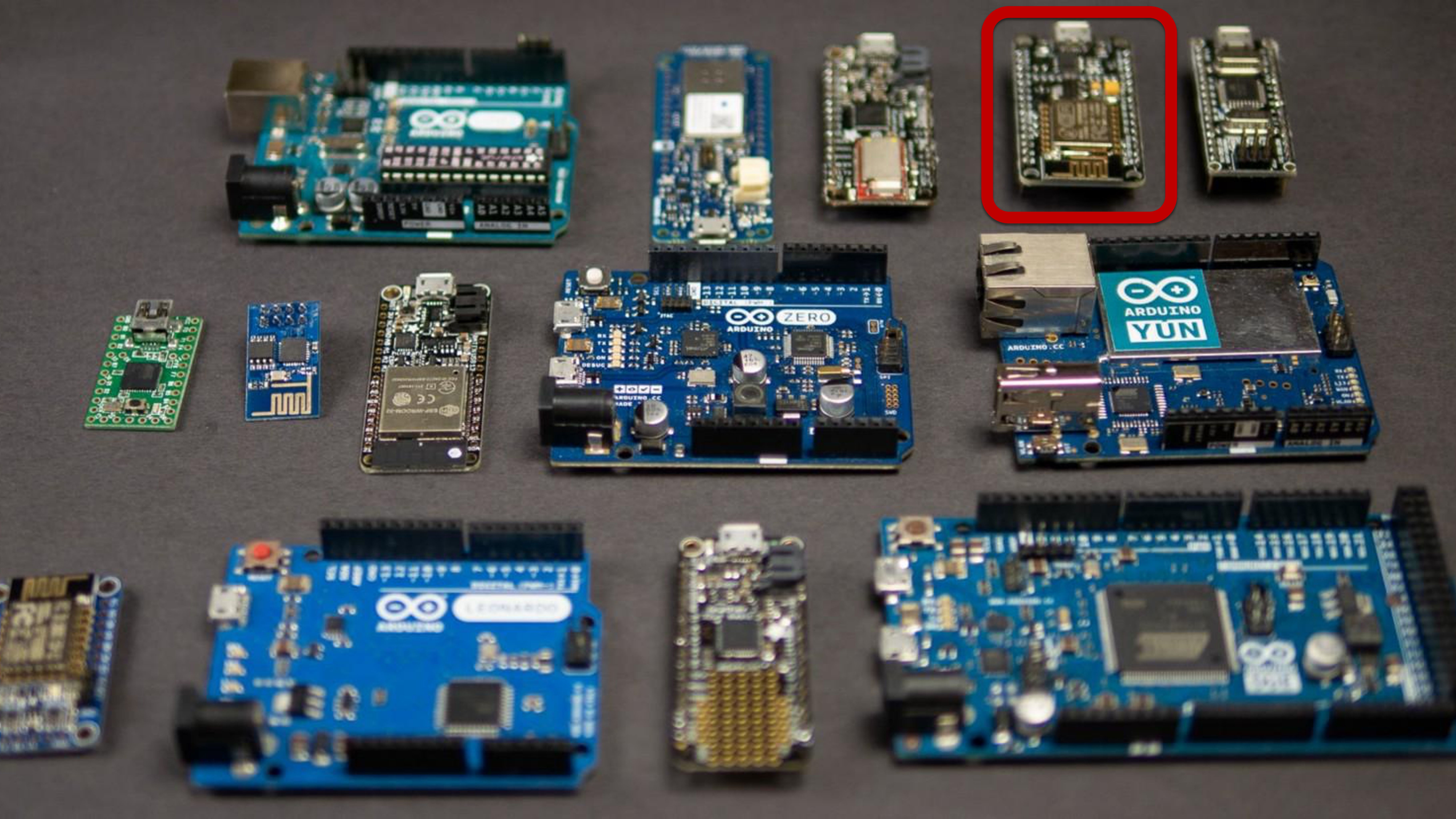
A desktop “claw”bot that can grab things
and cheer me up after every task I complete.

Questions?

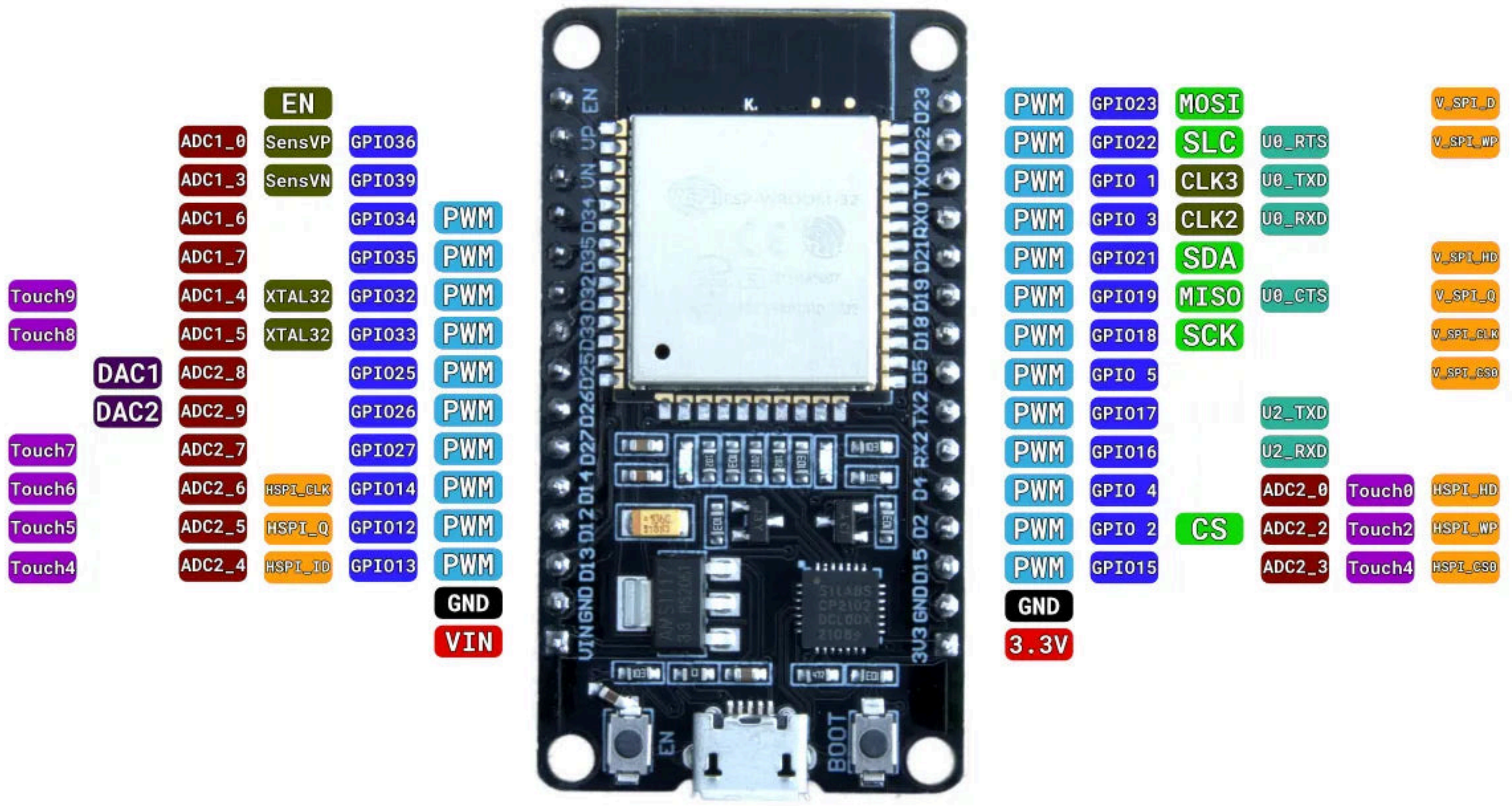
DigitalOutput

Huaishu Peng | UMD CS | Spring 2026




ZERO
ARDUINO
ARDUINO
YUN
ARDUINO.CC

What is Arduino?



ESP32 (30Pin version)

25 GPIO Pins
15 ADC-compactable pins
21 PWM
2 DAC
2 I2C

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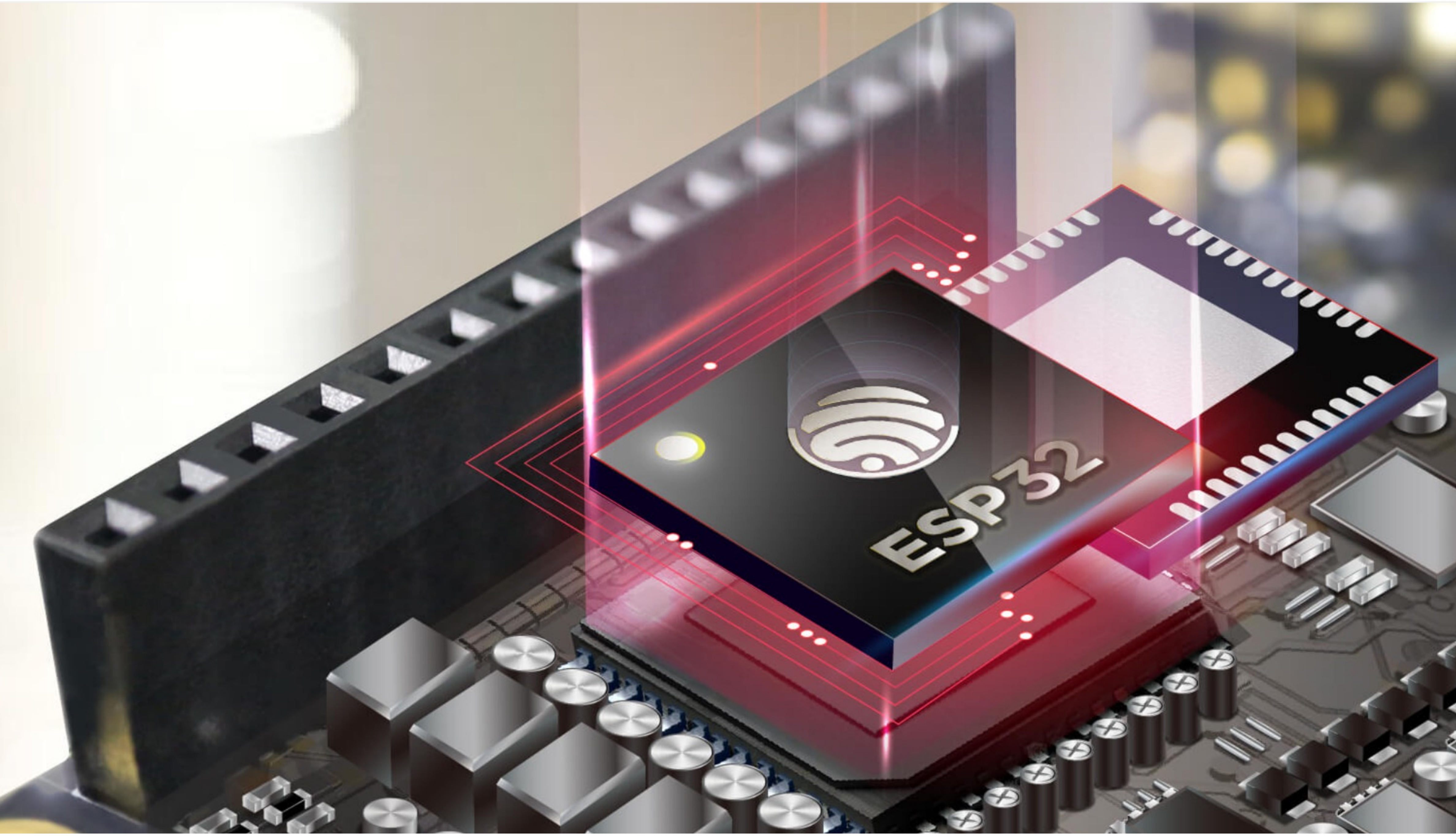
Hardware > ESP32 > ESP32

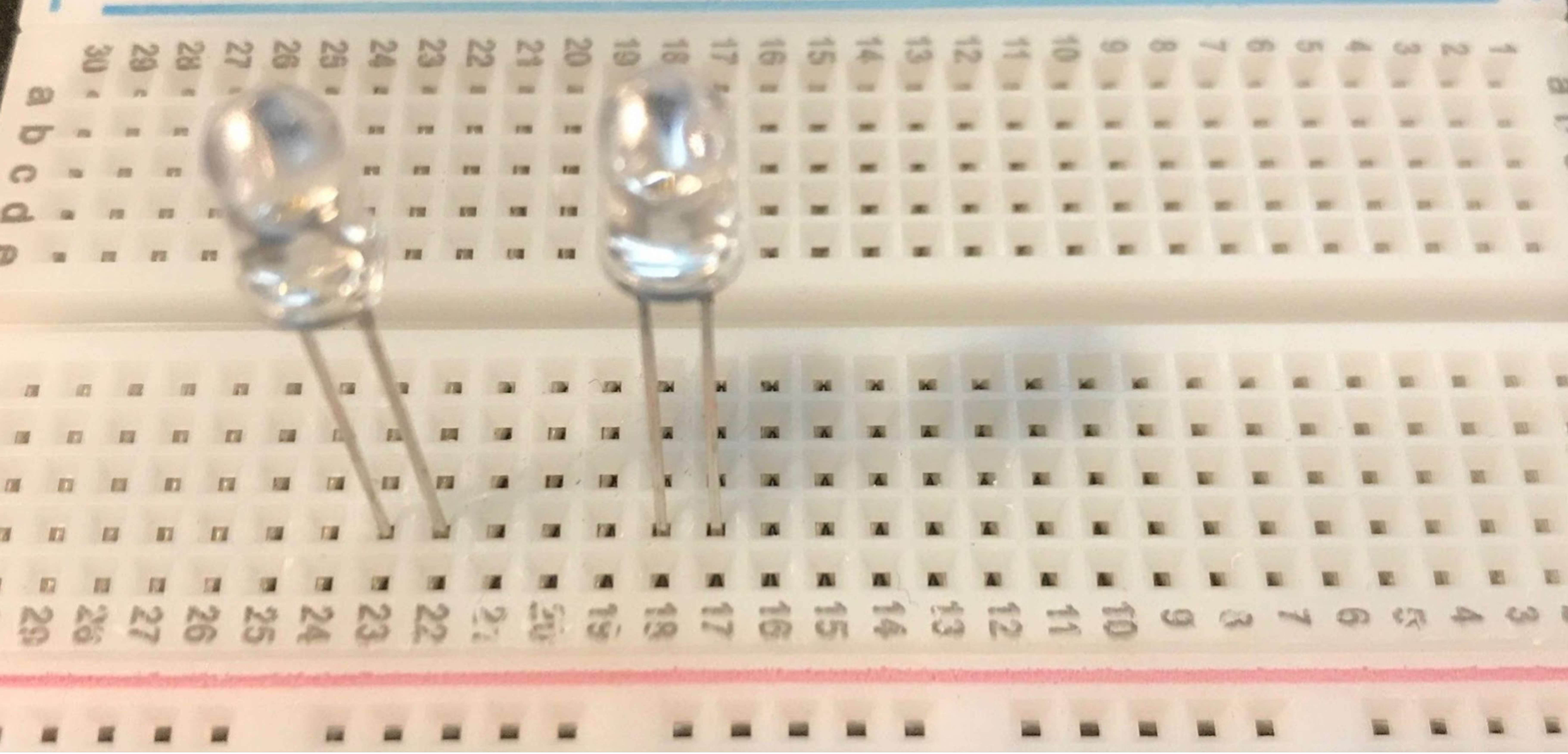
ESP32

A feature-rich MCU with integrated Wi-Fi and
Bluetooth connectivity for a wide-range
of applications

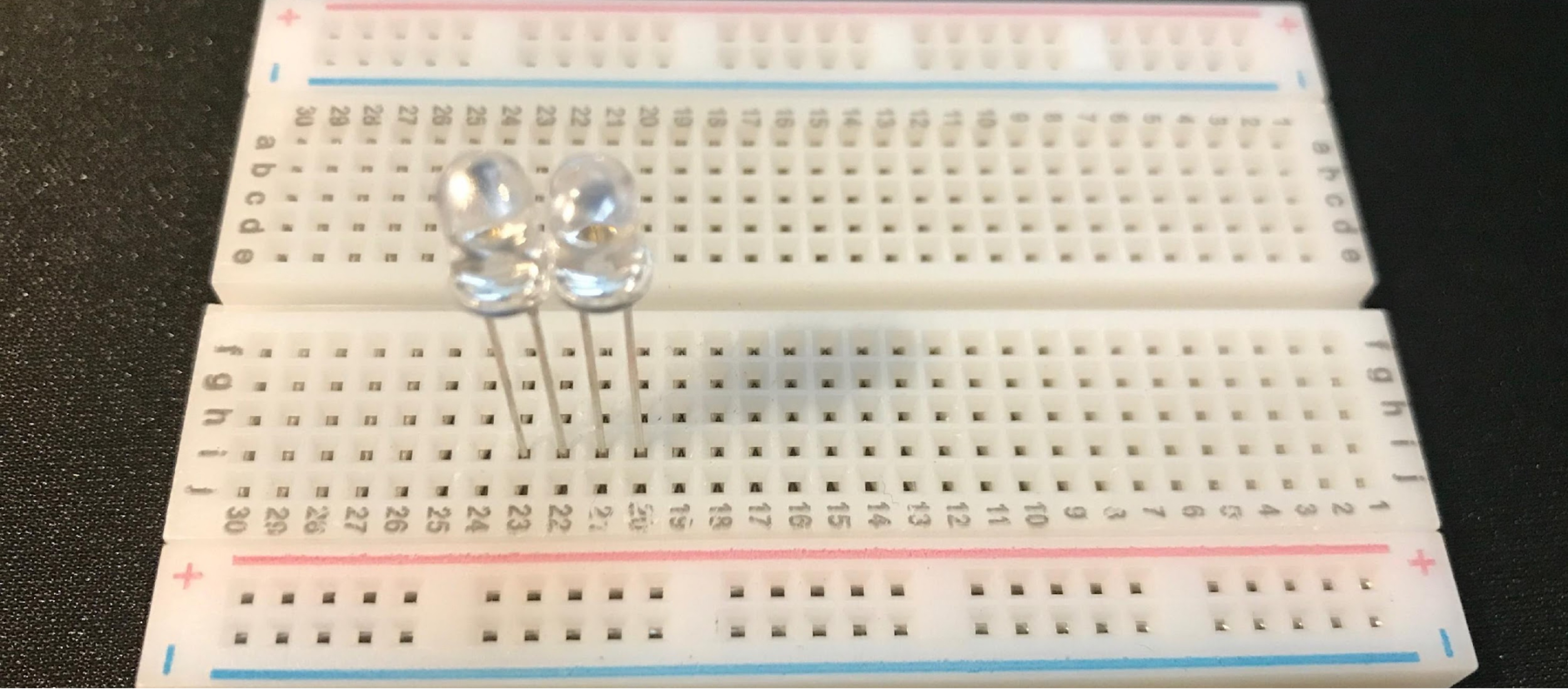


Datasheet: https://www.espressif.com/sites/default/files/documentation/esp32-wroom-32e_esp32-wroom-32ue_datasheet_en.pdf

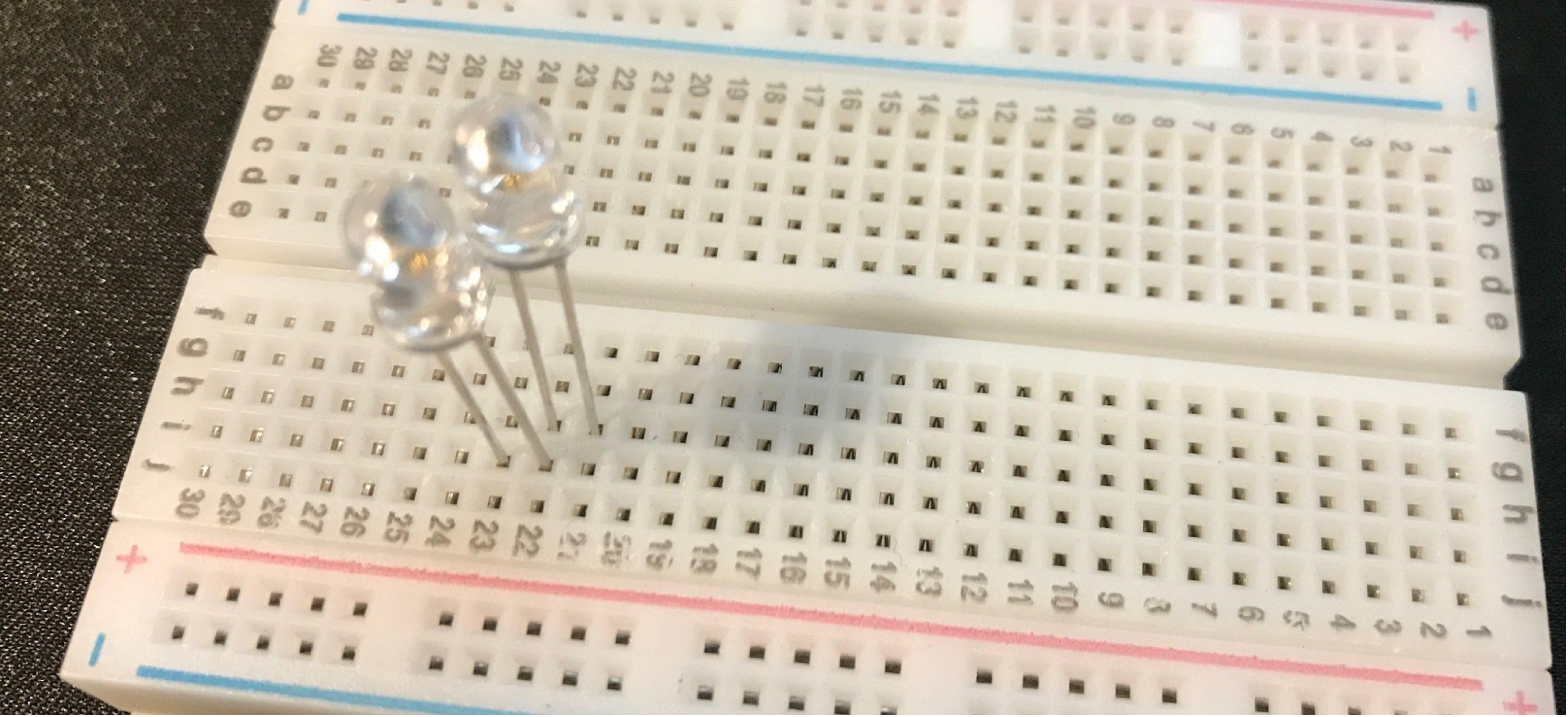

a b c d e
f g h i j
a b c d e
f g h i j



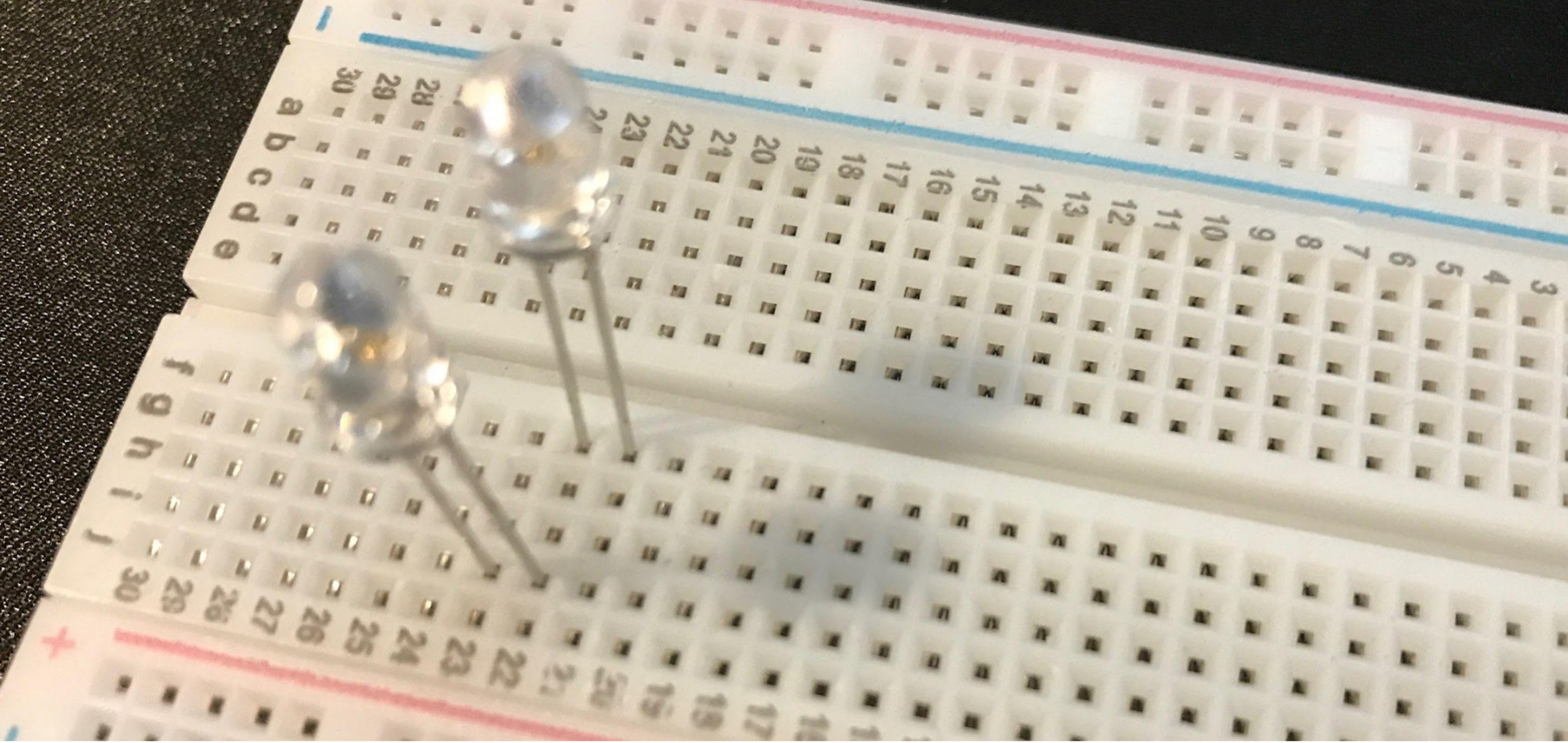
are the LEDs connected with each other?



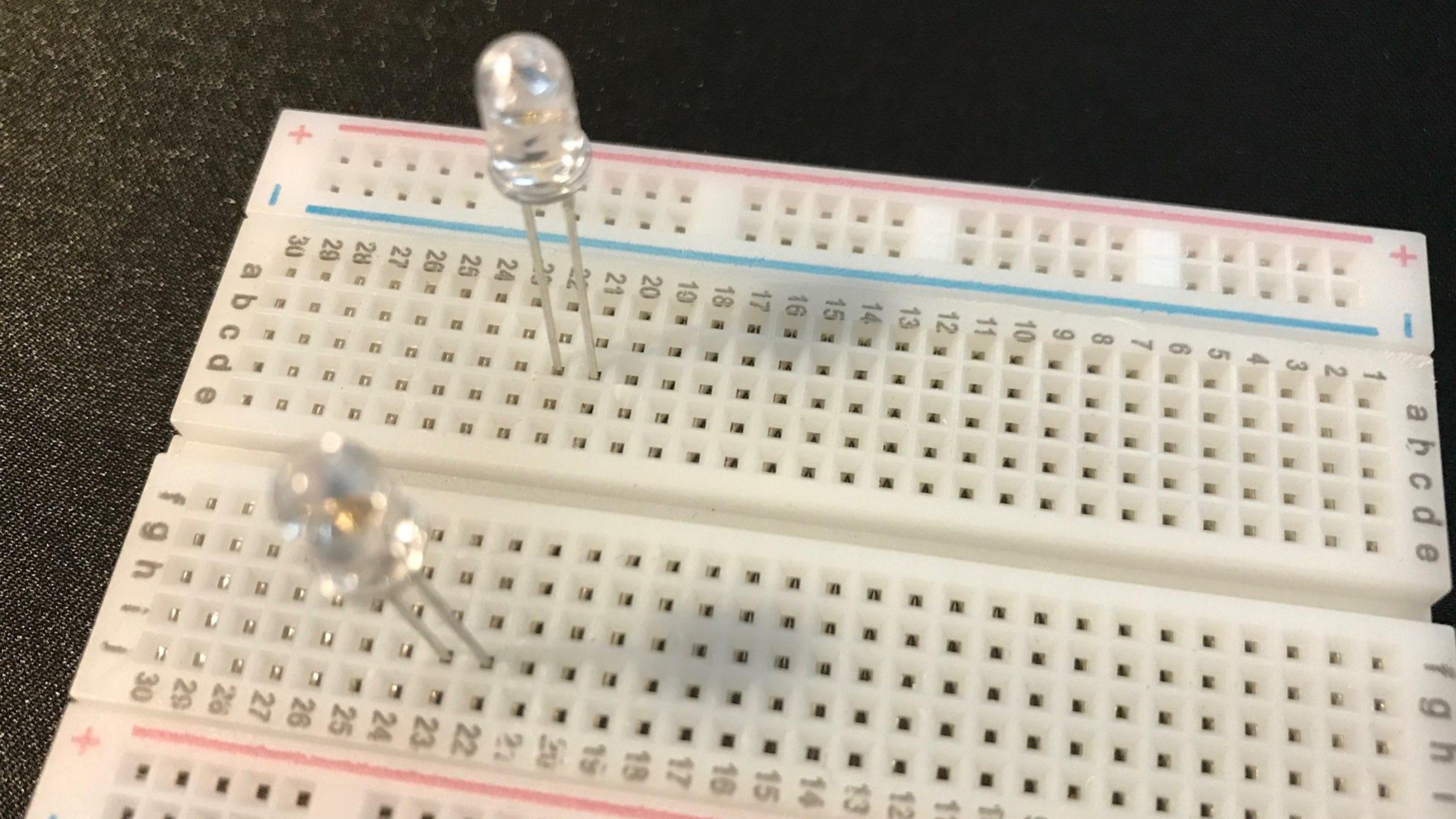
Now?

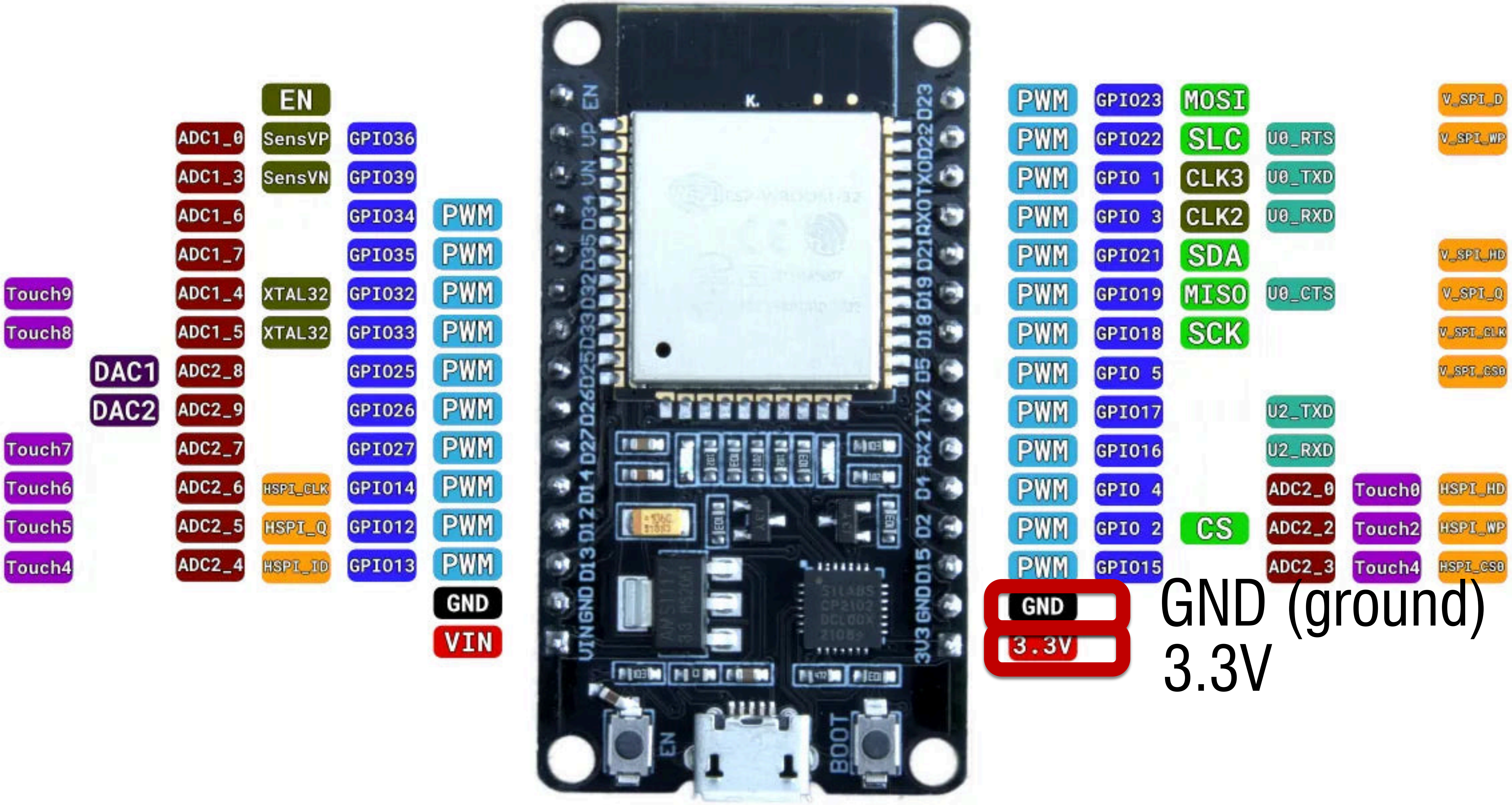


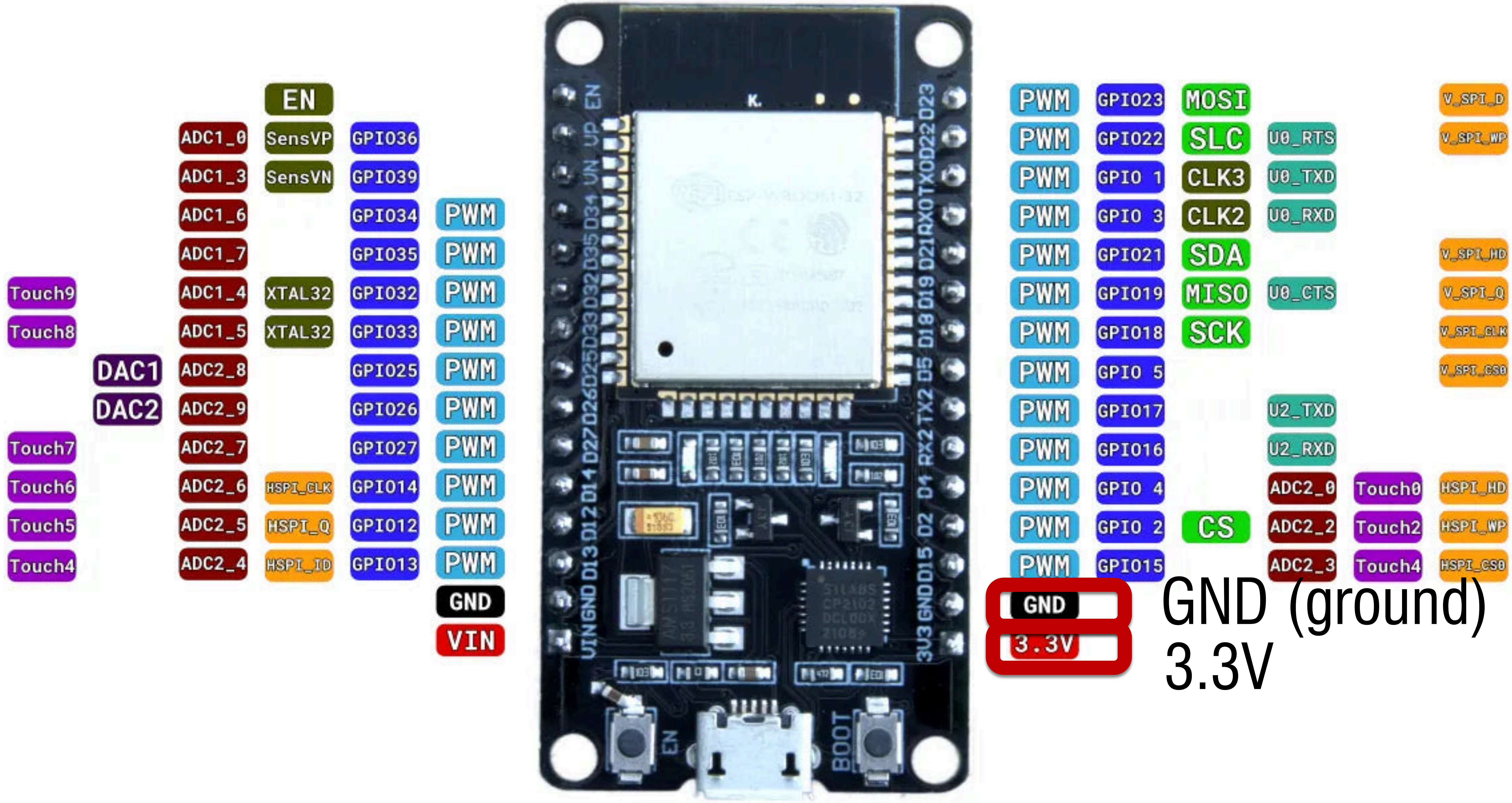
Now?


a b c d e
f g h i j
30 29 28 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3
30 27 26 25 24 23 22
+
-

Now?


+
-
30 29 28 27 26 25 24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1
a b c d e
f g h i j


EN
ADC1_0 SensVP GPIO36
ADC1_3 SensVN GPIO39
ADC1_6 GPIO34 PWM
ADC1_7 GPIO35 PWM
Touch9 ADC1_4 XTAL32 GPIO32 PWM
Touch8 ADC1_5 XTAL32 GPIO33 PWM
DAC1 ADC2_8 GPIO25 PWM
DAC2 ADC2_9 GPIO26 PWM
Touch7 ADC2_7 GPIO27 PWM
Touch6 ADC2_6 HSPI_CLK GPIO14 PWM
Touch5 ADC2_5 HSPI_Q GPIO12 PWM
Touch4 ADC2_4 HSPI_ID GPIO13 PWM
GND
VIN
PWM GPIO23 MOSI V_SPI_D
PWM GPIO22 SLC U0_RTS V_SPI_WP
PWM GPIO 1 CLK3 U0_TXD
PWM GPIO 3 CLK2 U0_RXD
PWM GPIO21 SDA V_SPI_HD
PWM GPIO19 MISO U0_CTS V_SPI_Q
PWM GPIO18 SCK V_SPI_CLK
PWM GPIO 5 V_SPI_CS0
PWM GPIO17 U2_TXD
PWM GPIO16 U2_RXD
PWM GPIO 4 ADC2_0 Touch0 HSPI_HD
PWM GPIO 2 CS ADC2_2 Touch2 HSPI_WP
PWM GPIO15 ADC2_3 Touch4 HSPI_CS0
GND
3.3V
GND (ground)
3.3V
EN
BOOT


EN
ADC1_0 SensVP GPIO36
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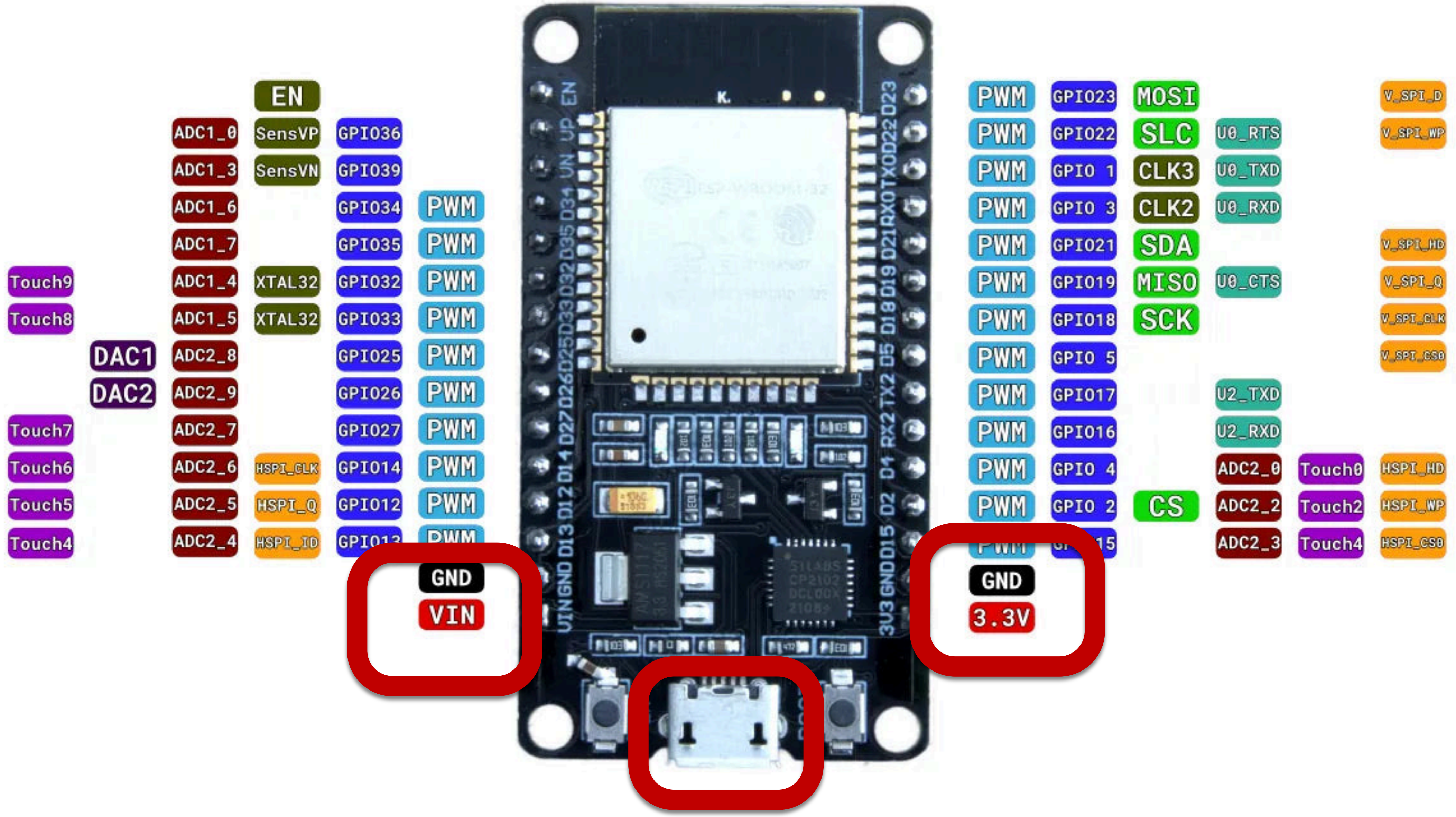
Touch8 ADC1_5 XTAL32 GPIO33 PWM
DAC1 ADC2_8 GPIO25 PWM
DAC2 ADC2_9 GPIO26 PWM
Touch7 ADC2_7 GPIO27 PWM
Touch6 ADC2_6 HSPI_CLK GPIO14 PWM
Touch5 ADC2_5 HSPI_Q GPIO12 PWM
Touch4 ADC2_4 HSPI_ID GPIO13 PWM
GND
VIN
PWM GPIO23 MOSI V_SPI_D
PWM GPIO22 SLC U0_RTS V_SPI_WP
PWM GPIO 1 CLK3 U0_TXD
PWM GPIO 3 CLK2 U0_RXD
PWM GPIO21 SDA V_SPI_HD
PWM GPIO19 MISO U0_CTS V_SPI_Q
PWM GPIO18 SCK V_SPI_CLK
PWM GPIO 5 V_SPI_CS0
PWM GPIO17 U2_TXD
PWM GPIO16 U2_RXD
PWM GPIO 4 ADC2_0 Touch0 HSPI_HD
PWM GPIO 2 CS ADC2_2 Touch2 HSPI_WP
PWM GPIO15 ADC2_3 Touch4 HSPI_CS0
GND
3.3V
GND (ground)
3.3V



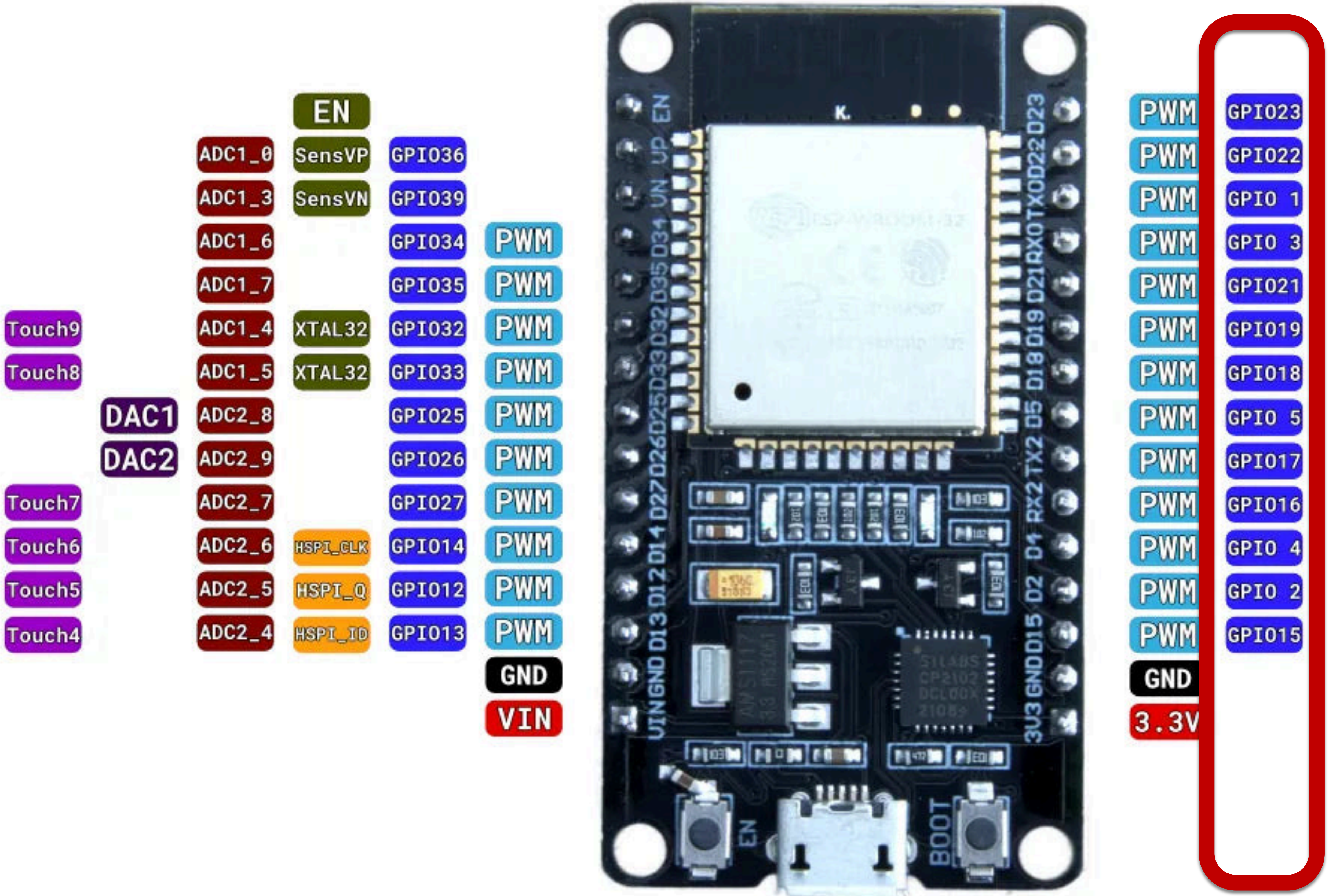
3 options to power up ESP32.

1. Directly via USB port.

2. Unregulated power to GND and VIN pins (Between 5 to 12 v, but 5 if possible)

3. Regulated power to GND and 3.3V pins (ONLY 3.3v!)

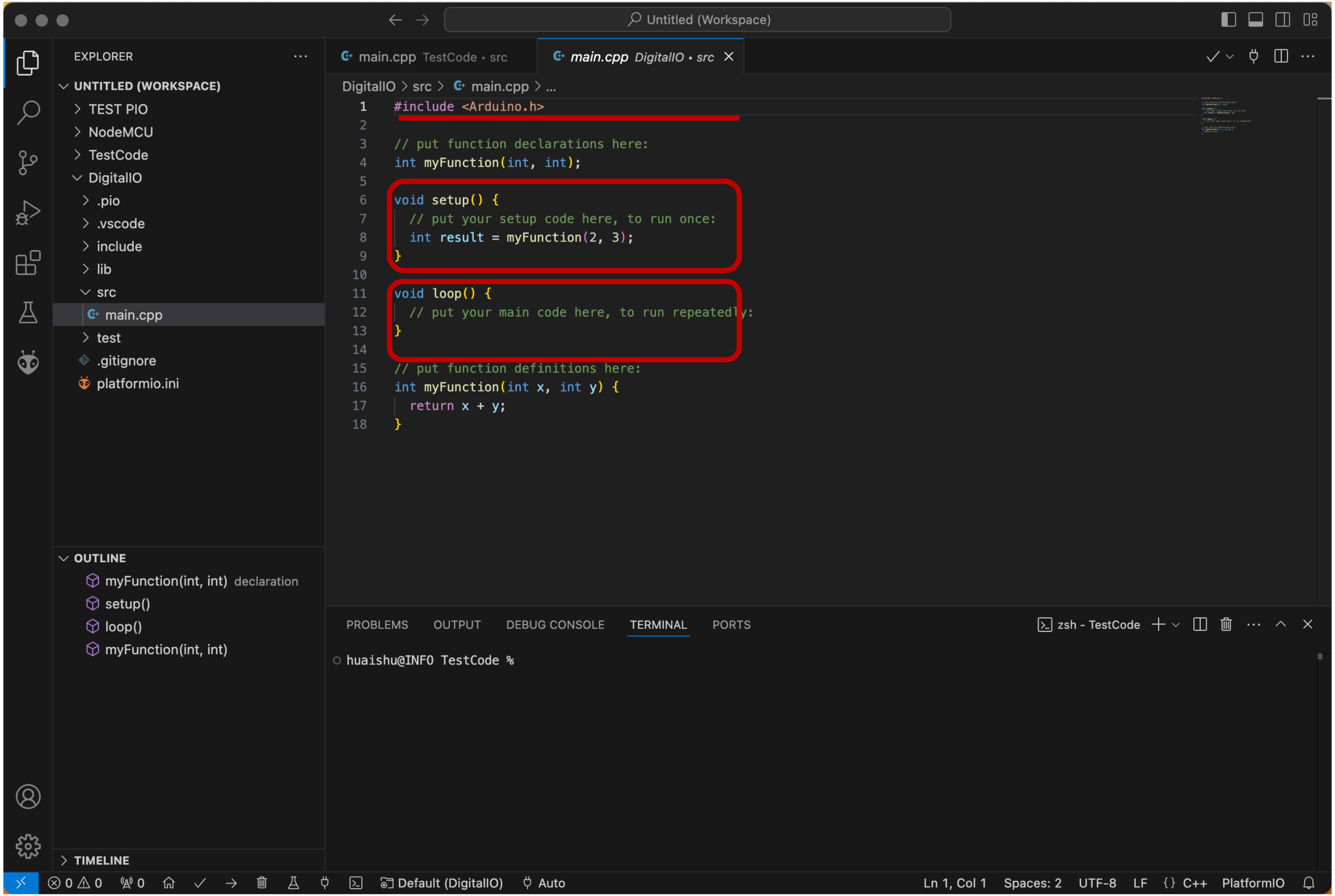
Always only power the microcontroller with one option



GPIO (General Purpose IO) for both digital input and output

Always refer to the pin layout

Arduino Framework Basics


Untitled (Workspace)
EXPLORER
UNTITLED (WORKSPACE)
TEST PIO
NodeMCU
TestCode
DigitalIO
.pio
.vscode
include
lib
src
main.cpp
test
.gitignore
platformio.ini
OUTLINE
myFunction(int, int) declaration
setup()
loop()
myFunction(int, int)
TIMELINE
main.cpp TestCode • src
main.cpp DigitalIO • src
DigitalIO > src > main.cpp > ...
#include <Arduino.h>
// put function declarations here:
int myFunction(int, int);
void setup() {
// put your setup code here, to run once:
int result = myFunction(2, 3);
}
void loop() {
// put your main code here, to run repeatedly:
}
// put function definitions here:
int myFunction(int x, int y) {
return x + y;
}
PROBLEMS
OUTPUT
DEBUG CONSOLE
TERMINAL
PORTS
zsh - TestCode
huaishu@INFO TestCode %
Default (DigitalIO)
Auto
Ln 1, Col 1
Spaces: 2
UTF-8
LF
C++
PlatformIO

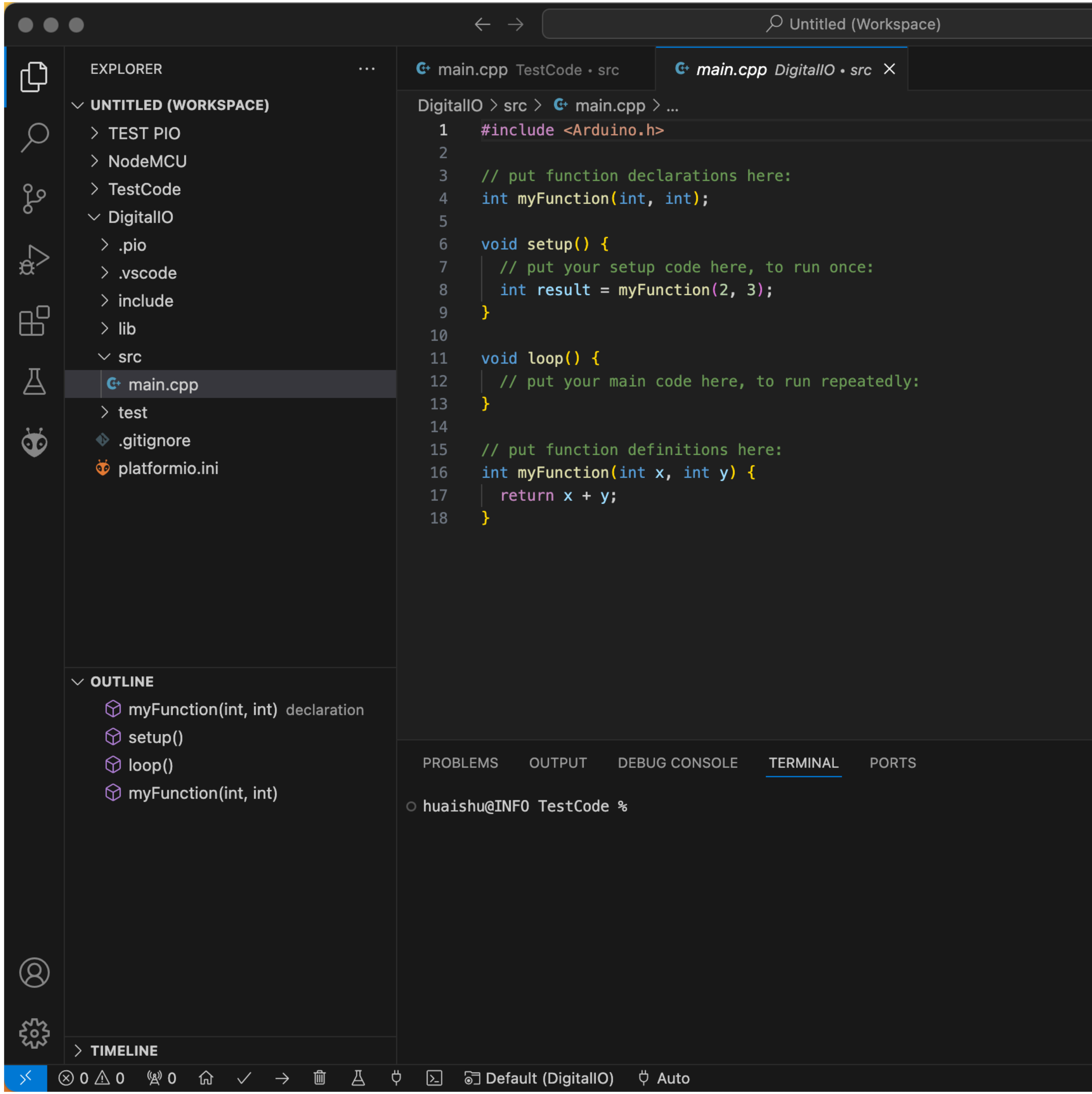
Digital Output – Blink an LED

Digital Output

Set the logic value of a pin
– LOW (0V) or HIGH (3.3V)

Arduino functions
– pinMode(pin, OUTPUT) to set the pin direction
Often in the setup() function
– digitalWrite(pin, value) to write the current value of a pin

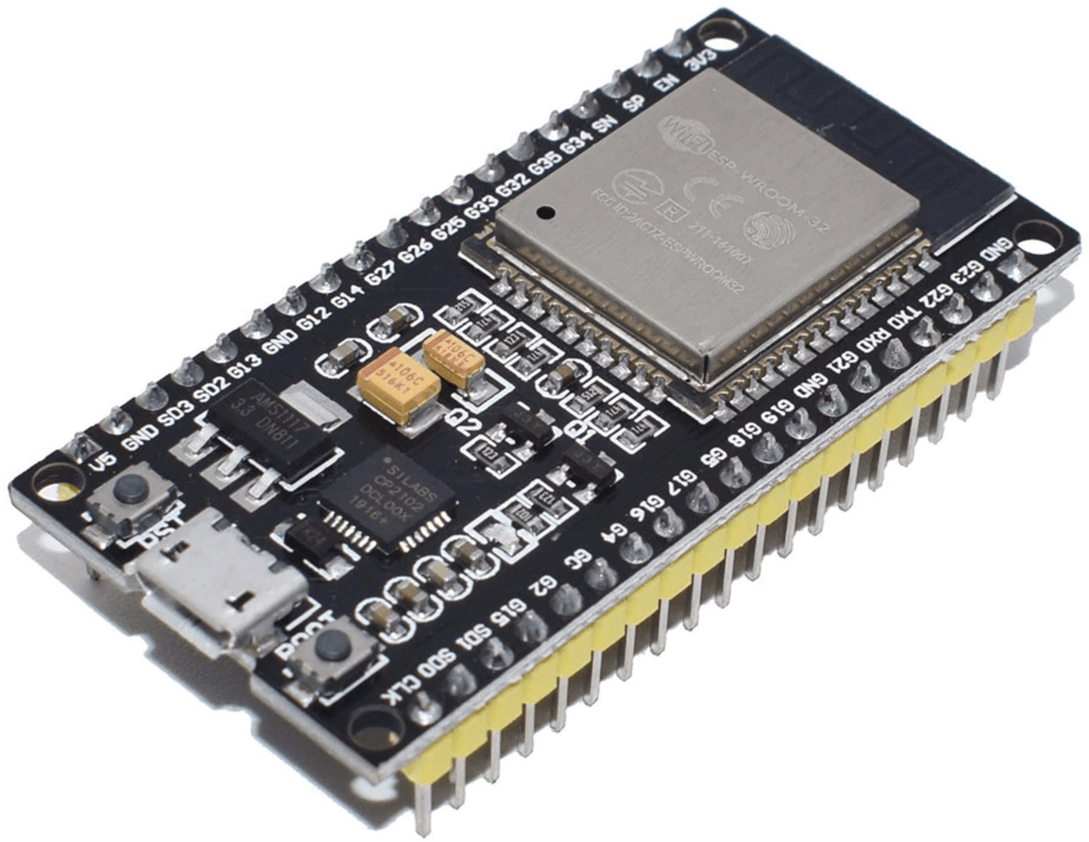
Limitations
– Only 0 or 3.3 V with limited current;



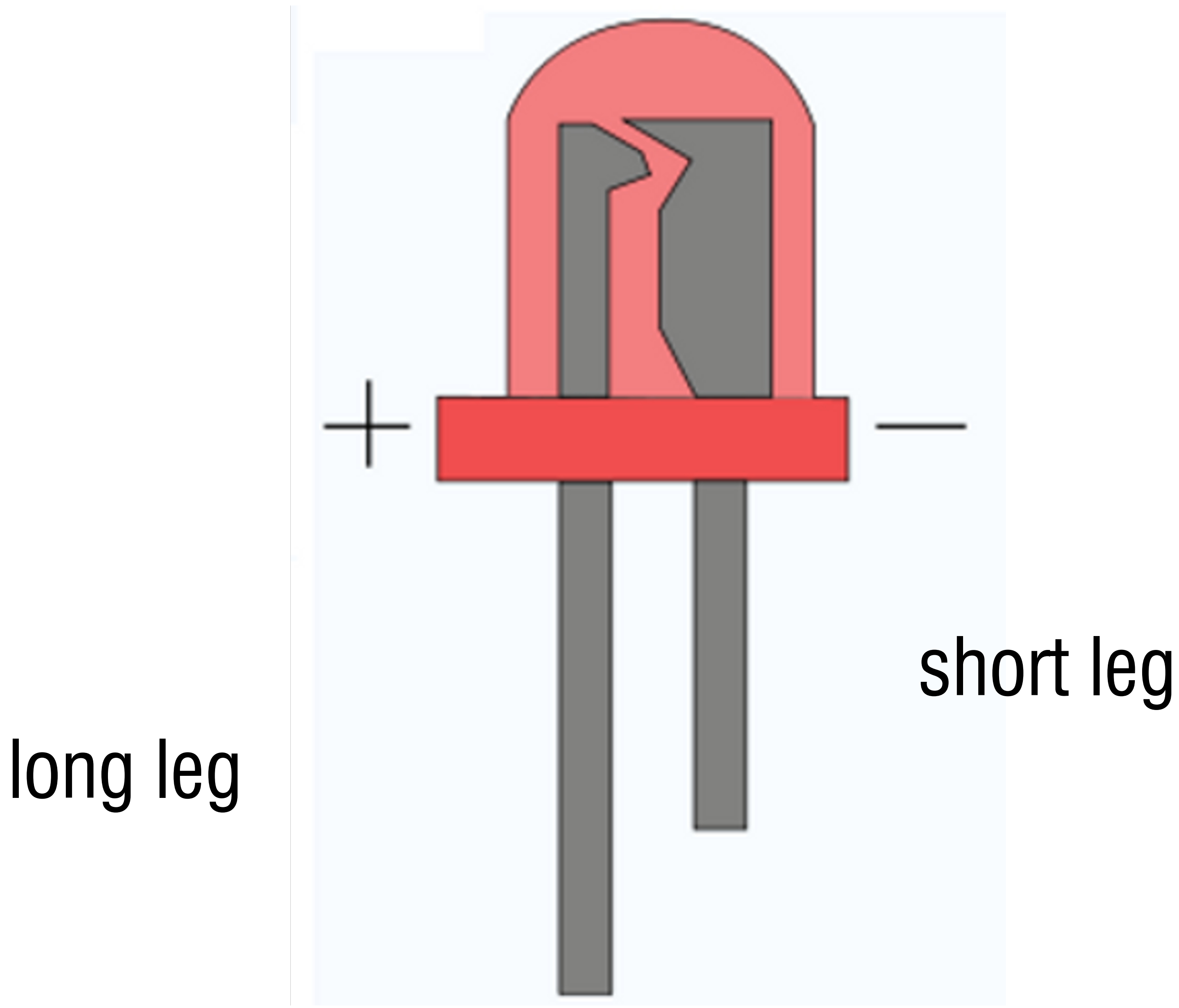
Blink the built-in LED

```
#include <Arduino.h>

// constants definition
const int ledPin = 2; // Default LED is connected to GPIO 2
// The setup() method runs once, when the sketch starts
void setup() {
  // initialize the digital pin as an output:
  pinMode(ledPin, OUTPUT);
}
// the loop() method runs over and over again,
// as long as the Arduino has power
void loop()
{
  digitalWrite(ledPin, HIGH); // set the LED on
  delay(5000); // wait for 5 second
  digitalWrite(ledPin, LOW); // set the LED off
  delay(5000); // wait for 5 second
}
```



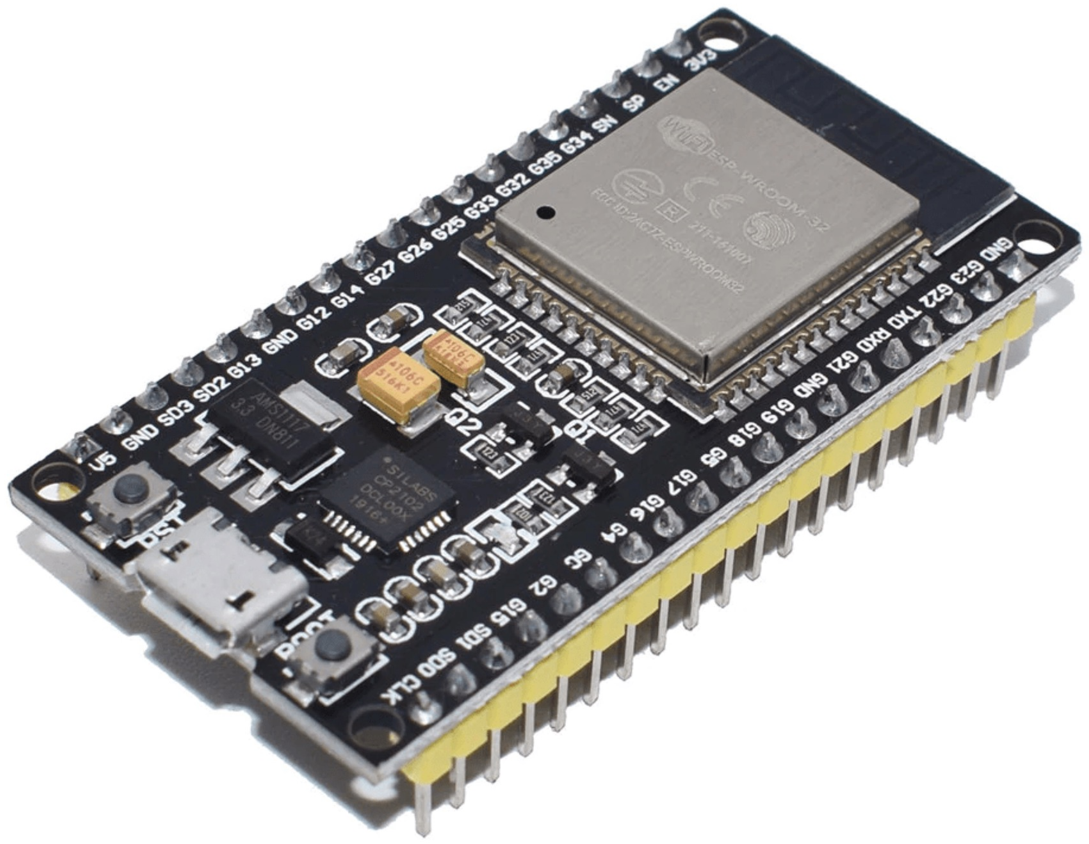
Practice: Light up an RED Led


+
−
short leg
long leg

Blink an external LED

```
#include <Arduino.h>

// constants definition
const int ledPin = 27; // Default LED is connected to GPIO 27
// The setup() method runs once, when the sketch starts
void setup() {
  // initialize the digital pin as an output:
  pinMode(ledPin, OUTPUT);
}
// the loop() method runs over and over again,
// as long as the Arduino has power
void loop()
{
  digitalWrite(ledPin, HIGH); // set the LED on
  delay(5000); // wait for 5 second
  digitalWrite(ledPin, LOW); // set the LED off
  delay(5000); // wait for 5 second
}
```



Serial Communication – talk to PC

Serial Communication

Setup
– Serial.begin(<baud_speed>)//9600

Receiving information
– Test is data is available
Serial.available()
– Read one byte
Serial.read()

Sending information
– Raw data transfer
Serial.write(val) or Serial.write(buf, len)

Other commands -> https://www.arduino.cc/reference/en

– Formatted output
Serial.print (x,{BIN,OCT,DEC,HEX})
– Read formatted data
Serial.parseFloat()
Serial.parseInt()

Echo program

```
// setup performs initializations
void setup()
{
  // initialize the serial port setting its speed to 9600 Baud:
  Serial.begin(9600);
}
// the loop() method runs over and over again,
// as long as the Arduino has power
void loop() {

if (Serial.available() > 0) {
   // Read the incoming byte
   char incomingByte = Serial.read();

   Serial.print("You entered: ");
   Serial.println(incomingByte);
  }
}
```

Assignment

Morse code

– Input:

3451 from the Serial Terminal Monitor

– Output:

Blink the LED accordingly

International Morse code

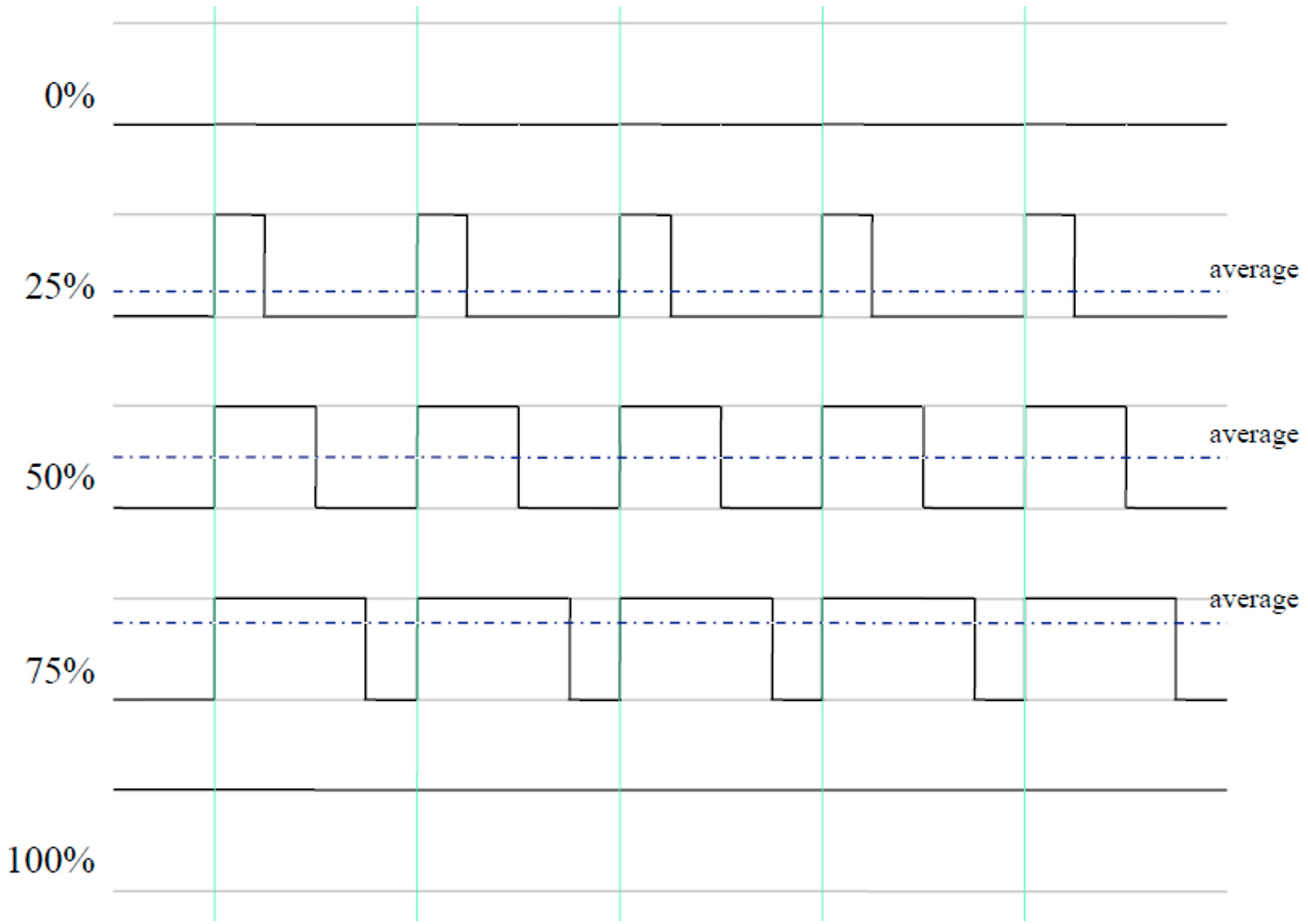
For this assignment:

A dot is 100ms long
A dash is equal to 3 dots
A space between parts of the same letter is equal to one dot
The space between two letters is equal to three dots

Submission:

Unlisted YouTube video link for the blinking LED
Upload the Arduino code

Pulse Width Modulation (PWM)



analogWrite() is on a scale of 0 - 255

Now modify your program to

blink the LED with 100% light intensity when type ‘1’ from the PC

turn it off when type ‘0’

light up with 50% light intensity when type ‘2’